that $M(G)$ has no self-loops by using the fact that if G is not cyclic, then G contains the central element $-\operatorname{Id} \in SU(2)$.

(d) Which groups from Problem 4.12.8 correspond to which diagrams?

(e) Using the McKay graph, find the dimensions of irreducible representations of all finite $G \subset SU(2)$ (namely, show that they are the numbers labeling the vertices of the affine Dynkin diagrams on our pictures). Compare with the results on subgroups of $SO(3)$ we obtained in Problem 4.12.8.

6.2. Indecomposable representations of the quivers A_1, A_2, A_3

We have seen that a central question about representations of quivers is whether a certain connected quiver has only finitely many indecomposable representations. In the previous subsection it is shown that only those quivers whose underlying undirected graph is a Dynkin diagram may have this property. To see if they actually do have this property, we first explicitly decompose representations of certain easy quivers.

Remark 6.2.1. By an object of the type $1 \longrightarrow 0$ we mean a map from a 1-dimensional vector space to the zero space. Similarly, an object of the type $0 \longrightarrow 1$ is a map from the zero space into a 1-dimensional space. The object $1 \longrightarrow 1$ means an isomorphism from a 1-dimensional to another 1-dimensional space. The numbers in such diagrams always mean the dimension of the attached spaces and the maps are the canonical maps (unless specified otherwise).

Example 6.2.2 (A_1)**.** The quiver A_1 consists of a single vertex and has no edges. Since a representation of this quiver is just a single vector space, the only indecomposable representation is the ground field (= a 1-dimensional space).

Example 6.2.3 (A_2)**.** The quiver A_2 consists of two vertices connected by a single edge:

$$\bullet \longrightarrow \bullet$$